

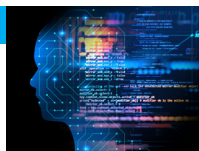
CONTENTS

TECHNOLOGY

- 22** TRENDS: Telecom Reforms In India And Futuristic 5G Applications
- 30** INTERVIEW: "Top Companies Need Highly Skilled Manpower And That Number Needs To Be Increased In India," — Prof. Debabrata Das, Director, IIIT Bangalore
- 36** FUTURE: Enhancing Defense With Quantum Computing And Intelligence
- 42** STANDARDS: Onboard IIoT Devices Seamlessly With The FDO Standard
- 46** SOFTWARE: Will ChatGPT Soon Replace Software Developers?

STRATEGY

- 65** TECH: AI For Everyone: Building Better Businesses With Low- Or No-Code AI



DESIGN

- 69** STRATEGY: Product Risk Analysis: Designing Robust Products
- 74** PCB: High-Speed PCB Design: Do It Right The First Time
- 78** SYSTEMS: Ultrasound On A Chip Supercharged With AI For Medical Scanning
- 80** AI: When To Use Free Or Open Solutions For AI
- 82** CIRCUIT: Reference Designs Of IoT Based Smart Plugs
- 84** IoT: Utilising Open Source For Cloud-Agnostic Delivery Of IoT Edge

DO-IT-YOURSELF

- 87** CIRCUIT IDEA: Kitchen Security Alarm Using MQ2 Sensor
- 89** PROJECT: Smart Ironing Device To Protect Your Clothes
- 93** PROJECT: Automatic Restricted Quantity Liquid Dispenser
- 96** PROJECT: Millisecond Clock Display Using ESP32 RTC
- 98** PROJECT: Anti-Collision System For Vehicles
- 102** PROJECT: Simple Water Temperature Indicator And Controller Using Arduino

REGULARS

- | | |
|--|---|
| 8 From You: Your Suggestions and Things You Wanted To Know! | 52 Dashboard: Key Industry Trends That Can Shape The Future Of Your Business |
| 10 Research & Innovation Updates | 54 Startups & Innovators |
| 15 Truly Innovative Electronics | 60 Industry News: Industry Updates |
| 17 New Products | |

EDITOR	: Rahul Chopra
EDITORIAL CORRESPONDENCE	: Phone: +91-9811155335 E-mail: editor@efy.in
SUBSCRIPTIONS & MISSING ISSUES	: Phone: +91-9811155335 E-mail: support@efy.in
BACK ISSUES, BOOKS, KITS, COMPONENTS, etc	: Phone: +91-9811155335 E-mail: support@efy.in
NEWSSTAND DISTRIBUTION	: Phone: +91-9811155335 E-mail: efycirc@efy.in
ADVERTISEMENTS	
NEW DELHI (HEAD OFFICE)	: Phone: +91-9811155335 E-mail: efyenq@efy.in
MUMBAI	: E-mail: rmowest@efy.in
BENGALURU	: E-mail: rmosouth@efy.in
CHINA	: Worldwide Focus Media E-mail: china@efy.in
GERMANY	: PMS Plantenberg Media Service GmbH E-mail: germany@efy.in
JAPAN	: Tandem Inc. E-mail: japan@efy.in
TAIWAN	: J.K. Media E-mail: taiwan@efy.in
UNITED KINGDOM	: ASA Media E-mail: uk@efy.in
UNITED STATES	: E & Tech Media E-mail: usa@efy.in

Printed, published and owned by Ramesh Chopra. Printed at M.P. Printers, B-220, Phase-2, NOIDA-201305, Uttar Pradesh, on the first day of each month and published from D-87/1, Okhla Industrial Area, Phase-1, New Delhi 110020. Copyright 2023. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any loss due to publishing errors. EFY will not be responsible for any wrong claims made by an advertiser. Disputes, if any, will be settled in a New Delhi court only.

SUBSCRIPTION RATES

EFY = Electronics For You

PRINT MAGAZINE	1 YEAR (12 copies each)	3 YEARS (36 copies each)	5 YEARS (60 copies each)
WITHIN INDIA (IN RUPEES)			
EFY (Rs 100/copy)	840	2150	3000
SAARC COUNTRIES (IN US\$ BY AIR)			
EFY	50	135	NA
OTHER COUNTRIES (IN US\$ BY AIR)			
EFY	100	270	NA

Courier delivery within India @ Rs 50 per copy

e-magazine subscriptions within India are available at half the rates mentioned here.

Overseas rates for each e-magazine in US\$:

1 year: \$12; 3 years: \$33; 5 years: \$50 only

Please send payments only in favour of EFY Enterprises Pvt Ltd through demand draft/cheque/money order to D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020, or online at: <http://subscribe.efyindia.com> or scan the QR code.

